LECTURE 4.3 / DIVISIBILITY

COMP233 • Number Theory & Proof Methods

ONE-PAGE KNOWLEDGE MAP

01 DIVISIBILITY: DEFINITION

Definition & equivalent condition

For integers $a \neq 0$ and b:

$a \mid b \Leftrightarrow \exists\, k \in \mathbb{Z} : b = ak$

Equivalently, b/a is an integer.

$3 \mid 21$: $21 = 3 \cdot 7$ | $-16 \mid 32$: $32 = (-16)(-2)$

$4 \nmid 15$: $15/4 \notin \mathbb{Z}$

02 SPECIAL & BOUNDARY CASES

For each nonzero a: $a \mid 0$, since $0 = a \cdot 0$.

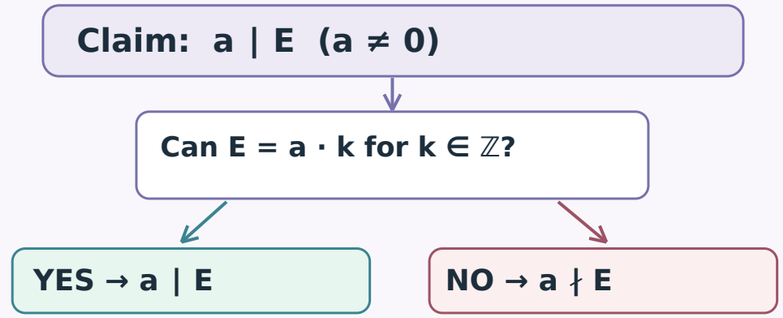
0 is not a permitted divisor (our convention).

Negative divisors are allowed: $7 \mid -7$.

“$a \mid b$” is a proposition; a/b is a number.

Do not infer $a = b$ from $a \mid b$.

03 PROOF DECISION ROUTE



04 DIRECT PROOFS & PROPERTIES

Proof by extracting an integer multiplier

$3a + 3b = 3(a + b)$, with $a + b \in \mathbb{Z}$

$10km = 5(2km)$, with $2km \in \mathbb{Z}$.

Transitivity: $a \mid b \wedge b \mid c \Rightarrow a \mid c$

$b = ar$, $c = bs$ $(r,s \in \mathbb{Z})$

$c = (ar)s = a(rs)$; $rs \in \mathbb{Z}$.

Therefore $a \mid c$.

05 DIVISOR BOUNDS & LOGIC

Positive divisor of positive integer

If $a,b > 0$ and $a \mid b$, then $a \leq b$.

Proof: $b = ak$, $k \in \mathbb{Z}$; $k \geq 1$, hence $b \geq a$.

More general, with $b \neq 0$

If $a \mid b$, $a > 0$ and $b \neq 0$, then $a \leq |b|$.

If $b = 0$, that bound need not hold.

All assumptions matter: $a > 0$, $b \neq 0$.

06 COUNTEREXAMPLES & REPAIR

False universal statement

$(a \mid b \wedge b \mid a) \Rightarrow a = b$ [FALSE]

Take $a = 2$, $b = -2$.

$2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$.

Correct replacement

$a \mid b$ and $b \mid a \Rightarrow |a| = |b|$.

If $a,b > 0$, then $a = b$.

07 PRIME DIVISIBILITY

A prime $p > 1$ has exactly two positive divisors:

1 and p. The number 1 is not prime.

Prime-divisor theorem

Every integer $n > 1$ has a prime divisor.
Proof idea: least counterexample m would be composite $m = uv$; a prime divisor of u would also divide m — contradiction.

08 UNIQUE PRIME FACTORIZATION

Each integer $n > 1$ is prime or has a unique product of prime factors up to reordering.

Standard factored forms

$72 = 2^3 \cdot 3^2$

$3300 = 2^2 \cdot 3 \cdot 5^2 \cdot 11$

09 APPLICATION: 17 DIVIDES m

Given the integer equality:

$8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot m =$

$17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10$

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17 is prime and divides the right product.
No displayed factor on the left contains 17.
Uniqueness forces $17 \mid m$.

BRIDGE TO §4.4 Divisibility is the remainder-zero case of $n = dq + r$, where $d > 0$ and $0 \leq r < d$.

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